Circuits (Capacitance and Resistance)

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1 Introduction

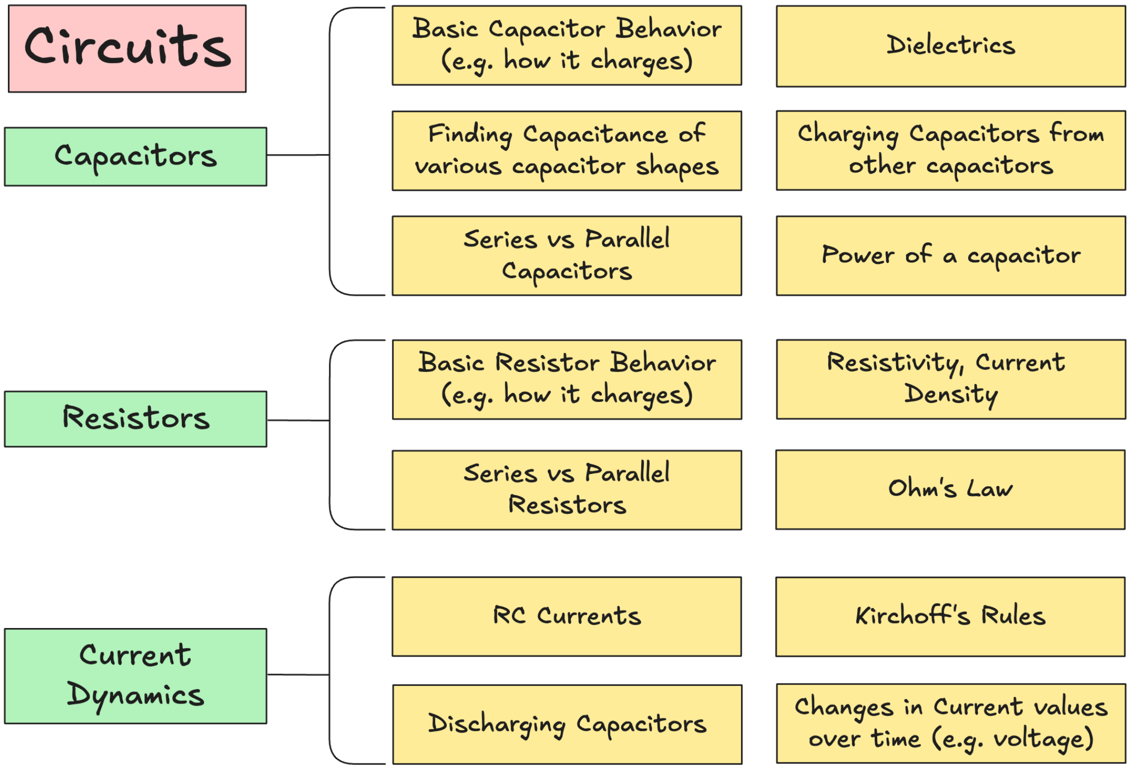

Figure 1: Circuitry Topics List

1.1 Basic Intuition for Charge Dynamics

- **Batteries**: Batteries do NOT supply charges (electrons come from capacitors / wires). Batteries are charge pumps, providing a source of energy.
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2 Capacitors

2.1 Basic Capacitor Behavior / Intuition

- Capacitors have the ability to store charge over a potential difference
- Capacitors are measured in F, or Farads. Unit Analysis: $\frac{[C]}{[V]} = \frac{[J]}{[C]} = [F]$

2.2 Finding the Capacitance of Capacitor Shapes

Basic capacitance is calculated by this formula:

$$C = \frac{\mathrm{q}}{\mathrm{V}}$$

Where q is the charge that the conductor has and the V is the potential difference across the capacitor.
Given that capacitance depends on the geometry between the plates, along with the medium between plates, one can calculate the capacitance of a capacitor with an odd geometry by:

1. Find the electric field between plates
2. Find the ΔV from the field
3. Apply that to the capacitance equation $C = \frac{q}{V}$

2.2.1 Finding the Electrical Field between two plates

With a area of A and a distance between of d, with the medium between them air (shown by the ϵ_0)

$$\Phi = EA = \frac{q_{\text{enc}}}{\epsilon_0} = \frac{q}{A\epsilon_0}$$
$$= \frac{\sigma}{\epsilon_0}$$

2.2.2 Finding voltage from field

$$\Delta V = -\int \vec{E}\delta\vec{s} = \int_{-\text{plate}}^{+\text{plate}} \vec{E}\delta\vec{s} = E\int \delta s = Ed$$

We removed the negative sign because while the field is going from the positive plate to the negative plate, the direction vector is doing the opposite. We also double the field because of the two plates.

2.2.3 Solving for capacitance

$$C = \frac{q}{V} = \frac{q}{\frac{q}{A\epsilon_0}} \times d = \frac{A\epsilon_0}{d}$$

2.3 Calculating for the Capacitance between Cylindrical Capacitors

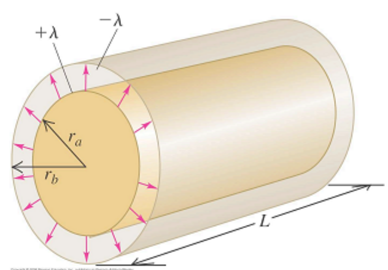

Figure 2: Examples of a Cylindrical Capacitor

2.3.1 Find Electrical Field between Two of the Cylinders

$$\Phi = EA = \frac{q_{\text{enc}}}{\epsilon_0}$$
$$A = 2\pi rL$$
$$E = \frac{q}{2\pi\epsilon_0 rL}$$

2.3.2 Find Voltage from Field

$$\Delta v = -\int \vec{E}\delta\vec{s} = -\int_b^a \frac{q}{2\pi rL\epsilon_0}\delta r = \int_a^b \frac{q}{2\pi rL\epsilon_0}\delta r = \frac{q}{2\pi L\epsilon_0}\ln\frac{b}{a}$$

2.3.3 Solving for the Equation

$$C = \frac{q}{v} = \boxed{\frac{2\pi L\epsilon_0}{\ln\frac{b}{a}}}$$

2.4 Series and Parallel Capacitors

Always remember the formula $C = \frac{Q}{V}$

2.4.1 Capacitors in Parallel

If a capacitor is in parallel, you can think it as adding up metal plates. Voltages are the same, and charge is proportionate to the capacitance at the capacitor. For an example 3 capacitors in series, the,

$$q_T = V_b \Sigma C$$
$$q_T = q_1 + q_2 + q_3$$
$$V_1 = V_2 = V_3 = V_B$$
$$C_T = C_{\text{eq}} = \frac{q_T}{V_T} = \frac{q_1}{V_1} + \frac{q_2}{V_2} + \frac{q_3}{V_3}$$
$$C_{\text{eq}} = \Sigma C$$

Note: This is opposite than resistors.

2.4.2 Capacitors in Series

If a capacitor is in series, you can think about it as adding up the air gaps. In the end, it's one big capacitor. Thus charge is equal.

$$q_T = q_1 = q_2 = q_3$$
$$V_T = V_1 + V_2 + V_3$$
$$\frac{1}{C_{\text{eq}}} = \frac{V_T}{q_T} = \frac{V_1 + V_2 + V_3}{q} = \frac{V_1}{q_1} + \frac{V_2}{q_2} + \frac{V_3}{q_3} = \frac{1}{C_1} + \frac{1}{C_2} = \frac{1}{C_3}$$

2.5 Dielectrics

Dielectrics

2.6 Charging capacitors from other capacitors

2.7 Power/Energy of a capacitors

One would think that since $q = CV_0$, and since $U = qV$, capacitance is CV^2. However, this is not true, since the voltage of a graph changes over time.

$$\Delta U = \int dq \delta U = \int V deltaq = \int \frac{q}{c} \delta q$$
$$= \frac{q^2}{2c}$$

3 Resistors

3.1 Basic Resistor Behavior / Intuition

3.2 Series vs Parallel Resistors

3.3 Resistivity, Current Density

3.3.1 Energy Density

Energy density is literally just the energy divided by the volume.

$$u = \frac{U}{\text{vol}} = \frac{\epsilon_0 V^2}{2d^2} = \frac{1}{2}\epsilon E^2$$

3.3.2 Current Density

J $= \frac{\delta i}{\delta A} = \int J \delta A$
The speed of electrons in a wire is VERY SLOW, the quantity of electrons give it its power.

3.3.3 Current Density

4 Resistance

You can calculate for the resistance of a resistor, given the formula $J = I/A$

$$\rho = \frac{E}{J} = \frac{V/L}{i/A} = \frac{V}{i}\frac{A}{L} = R\frac{A}{L}$$

$$R = \frac{\rho L}{A}$$

4.0.1 Electrical Power

$$P = \frac{\delta w}{\delta t} = \delta q V = i \delta t V$$

$$P = iV$$

4.1 Ohm's Law

Ohm's law relates voltage to current.
$I = \frac{V}{R}$

5 Current Dynamics

5.0.1 Current

Current is the measure of the flow of positive charges through a wire (which is basically the inverse of electrons, thanks Ben Franklin).
Current = i = $\frac{\delta q}{\delta t}$
Units: [i] = $\frac{C}{S}$ = A (Amperes).
Current going inside must be the same going outside, e.g. it is conserved.

5.1 Intuition about Currents

5.2 RC Currents (Resistor + Capacitor)

5.3 Kirchoff's Rules

5.4 Changes in Circuit Values over Time